

FRONTDESK OS — COMPLETE PRODUCT INTERFACE, DASHBOARD AND GITHUB REPOSITORY

You are an autonomous senior product designer, SaaS UX architect and frontend engineer.

You are not producing a concept, design description, wireframe or partial demo.

You must inspect the reference website, extract its visual system and implement a complete, production-quality interface for an omnichannel AI front desk application.

Do not ask me questions.

Do not stop after making the homepage.

Do not explain what you intend to do before doing it.

Inspect the existing files and environment, make reasonable product decisions and execute the complete specification.

1. OWNERSHIP AND DESIGN PERMISSION

The following website belongs to me:

`https://saffron-steam-experience.vercel.app/`

You have explicit permission to study, reproduce and reuse its visual design system for this project.

Use it as the absolute visual source of truth.

You may reproduce its:

- Typography treatment.
- Font scale.
- Spacing rhythm.
- Page composition.
- Grid system.
- Navigation treatment.
- Background treatment.
- Border styling.
- Card styling.
- Button styling.
- Corner radii.
- Shadows.

- Image framing.
- Section transitions.
- Hover states.
- Motion language.
- Responsive behaviour.
- Overall premium editorial feeling.

Do not reuse the café copy, café imagery, café icons or café business content.

Translate its visual language into a premium AI SaaS product.

The new application must feel as though the same elite design studio created both products.

2. REFERENCE AUDIT BEFORE IMPLEMENTATION

Before writing the new interface:

1. Open the reference website in a real browser.
2. Wait for animations and fonts to load.
3. Capture full-page screenshots at approximately:
 4. 1440px desktop.
 5. 1024px tablet.
 6. 768px small tablet.
 7. 390px mobile.
8. Inspect computed CSS values rather than guessing.
9. Record:
 10. Font families.
 11. Heading sizes.
 12. Body sizes.
 13. Line heights.
 14. Letter spacing.
 15. Container widths.
 16. Horizontal page padding.
 17. Section spacing.
 18. Main foreground and background colours.
 19. Accent colours.
 20. Border colours.
 21. Radii.
 22. Shadow values.
 23. Navigation dimensions.
 24. Button dimensions.
 25. Animation durations.
 26. Easing curves.

27. Convert the extracted visual values into reusable design tokens.
28. Use those tokens consistently across both the marketing website and application dashboard.

Do not create a generic blue-and-purple AI dashboard.

Do not create a standard shadcn starter interface.

Do not use random gradients, glowing blobs, excessive glassmorphism or cyberpunk styling.

The product should retain the warm, composed, premium and editorial visual character of the reference website.

3. PRODUCT IDENTITY

Use this working product identity:

Product name:

FrontDesk OS

Product category:

Omnichannel AI Front Desk

Primary positioning:

Every client conversation. One intelligent front desk.

Supporting message:

FrontDesk OS answers calls, manages email, responds on WhatsApp, handles website chat and books appointments—while keeping your team in control.

GitHub repository:

`witejackel-eng/ai-frontdesk-inbox`

GitHub repository description:

`Omnichannel AI front desk and shared client inbox for phone calls, email, WhatsApp, website chat and appointment scheduling.`

Use the GitHub identity:

`witejackel-eng`

Do not attribute commits to Z-AI, bots, assistants or generated authors.

Do not add AI-generated badges or AI authorship notices.

Treat FrontDesk OS as a working title. Structure the branding so the name can be changed later from one central configuration file.

4. PRODUCT OBJECTIVE

Build the complete interface for a SaaS application that allows a business to manage:

- Phone calls.
- Email.
- WhatsApp.
- Website live chat.
- Contacts.
- Leads.
- Appointments.
- AI receptionist behaviour.
- Knowledge sources.
- Automations.
- Human handoff.
- Team assignments.
- Analytics.
- Integrations.
- Billing.
- Security.
- Audit history.

The central product concept is:

A customer may begin through a phone call, continue over WhatsApp and later reply by email. FrontDesk OS must preserve those interactions as one continuous customer relationship.

The product must feel like a combination of:

- A premium shared inbox.
- An AI receptionist.
- A lightweight CRM.
- A call-management system.
- An appointment scheduler.
- An automation platform.
- A customer-intelligence dashboard.

Do not merely place unrelated modules next to each other.

Create one coherent operating system.

5. IMPLEMENTATION MODE

Build a complete frontend product prototype with:

- Fully implemented routes.
- Real navigation.
- Responsive layouts.
- Realistic seeded data.
- Working filters.
- Working search.
- Working drawers.
- Working modals.
- Working tabs.
- Working tables.
- Working dropdowns.
- Working composer states.
- Working call-control demonstrations.
- Working calendar interactions.
- Working AI configuration forms.
- Working automation builder interactions.
- Working onboarding.
- Loading, empty, error and disconnected states.

External provider integrations may use typed mock adapters unless valid credentials already exist.

Never pretend that an integration is live when it is not.

Disconnected integrations must clearly display:

- Demo mode.
- Connection required.
- Missing credentials.
- Simulated data.

Do not put fake production success messages in the product.

6. TECHNOLOGY

Use the latest stable releases that are compatible with one another.

Preferred foundation:

- Next.js App Router.
- React.
- TypeScript with strict mode.
- Tailwind CSS.
- Accessible Radix or shadcn primitives.
- Motion or Framer Motion for purposeful animation.

- TanStack Query where asynchronous state is needed.
- TanStack Table for complex tables.
- React Hook Form.
- Zod.
- date-fns.
- Recharts for analytics.
- Lucide icons.
- cmdk for command search.
- Sonner for notifications.
- dnd-kit where drag-and-drop improves workflow.
- A lightweight state store only where justified.

Use a package only when it solves a real product requirement.

Do not install duplicate libraries that perform the same job.

Do not add a component library and then leave its default appearance intact.

Every component must be visually adapted to the extracted reference design system.

Do not introduce a database requirement merely to display the frontend prototype.

Create a clean adapter architecture that can later connect to production APIs.

7. PROJECT ARCHITECTURE

Use a maintainable folder structure similar to:

```
src/  
  app/  
    (marketing)/  
    (auth)/  
  app/  
    onboarding/  
  api/  
components/  
  app-shell/  
  inbox/  
  calls/  
  contacts/  
  calendar/  
  ai-agent/  
  knowledge/  
  automations/  
  analytics/  
  integrations/  
  settings/  
  marketing/
```

```
shared/  
  ui/  
config/  
data/  
hooks/  
lib/  
  adapters/  
  ai/  
  channels/  
  permissions/  
  security/  
providers/  
stores/  
types/  
styles/  
tests/
```

Centralize:

- Product name.
- Product metadata.
- Navigation.
- Channel definitions.
- Feature flags.
- Design tokens.
- Demo workspace information.

Do not scatter product names and colour values across files.

8. REQUIRED ROUTES

Marketing and authentication

Create:

- /
- /product
- /channels
- /ai-front-desk
- /integrations
- /pricing
- /security
- /login
- /signup

Onboarding

Create:

- /onboarding
- /onboarding/business
- /onboarding/channels
- /onboarding/knowledge
- /onboarding/agent
- /onboarding/hours
- /onboarding/test
- /onboarding/complete

Product application

Create:

- /app
- /app/inbox
- /app/inbox/[conversationId]
- /app/calls
- /app/calls/[callId]
- /app/contacts
- /app/contacts/[contactId]
- /app/calendar
- /app/ai-agent
- /app/knowledge
- /app/automations
- /app/analytics
- /app/team
- /app/integrations
- /app/settings
- /app/settings/channels
- /app/settings/notifications
- /app/settings/security
- /app/billing
- /app/audit-log

Every visible navigation item must lead somewhere.

Do not create dead buttons or placeholder links.

9. MARKETING HOMEPAGE

The marketing website must reuse the reference site's design system while communicating a serious SaaS product.

Header

Include:

- FrontDesk OS wordmark.
- Product.
- Channels.
- AI Front Desk.
- Integrations.
- Pricing.
- Security.
- Sign in.
- Primary “Start building” button.

Create a polished responsive navigation and mobile menu.

Use the same header proportions, spacing and interaction quality as the reference site.

Hero

Primary headline:

Every client conversation. One intelligent front desk.

Supporting copy:

Answer phone calls, manage email, respond on WhatsApp, handle website chat and book appointments from one AI-powered workspace.

Primary CTA:

Explore the dashboard

Secondary CTA:

See how it works

The main hero visual must be a live product composition, not an unrelated stock image.

Create an animated interface preview that demonstrates:

1. An incoming phone call.
2. A live transcript.
3. The AI checking calendar availability.
4. An appointment being created.
5. A WhatsApp confirmation being sent.
6. The conversation appearing in the unified inbox.

Animation should remain restrained, readable and premium.

Homepage sections

Create these sections:

1. Channel overview.
2. Unified customer timeline.
3. AI receptionist capabilities.
4. Phone-call experience.
5. Appointment scheduling.
6. AI and human handoff.
7. Knowledge and business context.
8. Automation.
9. Analytics.
10. Security and control.
11. Integrations.
12. Pricing preview.
13. Final CTA.

Avoid vague AI buzzwords.

Use direct, believable product language.

Do not add invented customer logos.

Do not add fabricated testimonials.

Do not claim numerical outcomes that have not been proven.

Demo statistics must be clearly identified as demonstration data.

10. APPLICATION SHELL

Create a premium, responsive product shell.

Desktop sidebar

Include:

- Overview.
- Inbox.
- Calls.
- Contacts.
- Calendar.
- AI Front Desk.
- Knowledge.
- Automations.
- Analytics.
- Team.

- Integrations.
- Settings.

The sidebar should support:

- Collapsed state.
- Active route.
- Keyboard navigation.
- Workspace switcher.
- Channel-health indicator.
- User menu.

Top bar

Include:

- Contextual page title.
- Global search.
- Command palette shortcut.
- AI status.
- Notifications.
- Help.
- User avatar.

Mobile

Create a deliberate mobile product experience.

Do not squeeze the desktop sidebar onto a small screen.

Use:

- Mobile bottom navigation for primary destinations.
- Slide-over navigation for secondary areas.
- Full-screen conversation views.
- Touch-friendly controls.
- Sticky action areas where necessary.

11. OVERVIEW DASHBOARD

The overview must answer:

- What needs attention?
- What has the AI handled?
- Are any channels disconnected?
- Are appointments being booked?
- Where are clients waiting?
- Is the AI making mistakes?

- What should the team do next?

Include:

- Open conversations.
- Waiting for human.
- AI-resolved conversations.
- Missed calls recovered.
- Appointments booked.
- Qualified leads.
- Average first-response time.
- AI confidence trend.
- Channel volume.
- Upcoming appointments.
- Urgent conversations.
- Recent AI actions.
- Channel health.
- Team availability.

Use believable demonstration values.

Charts must be readable and purposeful.

Do not fill the page with unnecessary miniature cards.

Create hierarchy and breathing room.

12. UNIFIED INBOX

This is the centre of the product.

Use a polished three-panel desktop layout:

Left panel: navigation and filters

Include:

- All conversations.
- Unassigned.
- Mine.
- Waiting.
- AI handling.
- Needs approval.
- Snoozed.
- Closed.
- Spam.

Channel filters:

- Phone.
- Email.
- WhatsApp.
- Website chat.

Additional filters:

- Priority.
- Assignee.
- Intent.
- Sentiment.
- AI confidence.
- SLA status.
- Date.
- Tags.

Middle panel: conversation list

Each conversation should show:

- Contact avatar.
- Contact name.
- Company.
- Channel icon.
- Latest message or call summary.
- Timestamp.
- Unread state.
- Assignee.
- Priority.
- AI or human state.
- SLA indicator.
- Appointment or lead indicator.
- Tags.

Include:

- Search.
- Sorting.
- Bulk actions.
- Keyboard navigation.
- Density control.

Main panel: conversation

Support all channel types.

Include:

- Message timeline.

- Email threads.
- Call transcript blocks.
- WhatsApp delivery states.
- Website-chat events.
- Attachments.
- Internal notes.
- AI actions.
- Appointment events.
- Channel-switch events.
- Human takeover events.

The composer must support:

- Public reply.
- Internal note.
- Channel selection.
- Attachment.
- Saved reply.
- AI draft.
- Rewrite.
- Shorten.
- Change tone.
- Translate.
- Schedule send.
- Request approval.

Clearly distinguish:

- Customer message.
- AI message.
- Human message.
- Internal note.
- System event.

Right context panel

Include:

- Contact details.
- Connected identities.
- Company.
- Lead stage.
- Lifetime conversation history.
- Upcoming appointment.
- Previous appointments.
- Notes.
- Tags.
- Custom fields.
- Current AI summary.
- Suggested next action.
- Knowledge sources used.

- Related conversations.

The context panel should collapse when more reading space is required.

13. PHONE AND CALL EXPERIENCE

Create a complete phone-call interface.

Calls list

Display:

- Caller.
- Direction.
- Status.
- Duration.
- AI or human handler.
- Outcome.
- Sentiment.
- Appointment created.
- Lead status.
- Recording availability.
- Date and time.

Filters:

- Incoming.
- Outgoing.
- Missed.
- Voicemail.
- Transferred.
- AI handled.
- Human handled.
- Failed.

Live call workspace

Create a sophisticated live-call view with:

- Caller identity.
- Telephone number.
- Call timer.
- Audio waveform.
- AI listening indicator.
- AI speaking indicator.
- Live transcript.
- Current detected intent.
- Sentiment.
- Confidence.

- Customer record.
- Suggested responses.
- Calendar availability.
- Tool-action activity.
- Mute.
- Hold.
- Transfer.
- Add human.
- End call.
- Take over from AI.

Include a visible recording-consent state.

Completed call

Display:

- Recording player.
- Transcript.
- AI summary.
- Call outcome.
- Topics.
- Questions asked.
- Commitments made.
- Follow-up tasks.
- Appointment.
- Sentiment timeline.
- Quality flags.
- AI mistakes.
- Suggested coaching.

Use simulated call functionality where a real telephony provider is not connected.

Label it clearly as demo mode.

14. CONTACTS AND CLIENT CRM

Create a lightweight CRM designed around conversations.

Contacts list:

- Name.
- Company.
- Email.
- Telephone.
- Last interaction.
- Primary channel.
- Lead stage.
- Owner.

- Upcoming appointment.
- Sentiment.
- Tags.

Contact profile:

- Unified customer timeline.
- Channel identities.
- Contact information.
- Business information.
- Lead stage.
- Estimated value.
- Assigned owner.
- Notes.
- Files.
- Appointments.
- Conversations.
- Calls.
- AI-generated summary.
- Preferences.
- Consent history.
- Data export.
- Delete request status.

Allow contact merge and identity merge through safe confirmation dialogs.

15. CALENDAR AND SCHEDULING

Create:

- Day view.
- Week view.
- Agenda view.
- Team availability.
- Appointment details.
- Booking status.
- Appointment source.
- Assigned team member.
- Channel through which it was booked.
- Time-zone display.
- Buffer settings.
- Working hours.
- Blackout dates.
- Booking types.

Integrations should include:

- Google Calendar.
- Microsoft Calendar.
- Notion.

- Zoom.
- Google Meet.
- Microsoft Teams.

Architectural honesty:

- Use Google Calendar or Microsoft Calendar as the actual scheduling source in the interface model.
- Treat Notion Calendar as a connected view of the underlying calendar.
- Use Notion integration for client pages, notes, databases and tasks.
- Do not falsely represent Notion Calendar as a full server-side scheduling API.

Show integration status and sync health.

16. AI FRONT DESK BUILDER

This page must feel like the control centre for the AI receptionist.

Create sections for:

Identity

- Agent name.
- Role.
- Business name.
- Greeting.
- Closing.
- Tone.
- Formality.
- Languages.
- Pronunciation dictionary.

Behaviour

- Ask one question at a time.
- Confirm important details.
- Never invent pricing.
- Never promise unavailable times.
- Avoid restricted topics.
- Ask for human help when uncertain.
- Disclose that the customer is speaking with AI.
- Respect working hours.
- Handle interruptions.
- Handle silence.
- Handle angry customers.

Channel settings

Separate controls for:

- Phone.
- Email.
- WhatsApp.
- Website chat.

Allow different:

- Greetings.
- Response lengths.
- Waiting times.
- Disclosure text.
- Escalation rules.
- Business-hours behaviour.

Tool permissions

Create clear permission controls for:

- Read calendar availability.
- Book appointment.
- Reschedule appointment.
- Cancel appointment.
- Create contact.
- Update contact.
- Create lead.
- Send email.
- Send WhatsApp message.
- Transfer call.
- Create task.
- Apply tags.

Each permission should support:

- Disabled.
- AI may suggest.
- Requires approval.
- AI may execute.

Confidence and handoff

Allow configuration of:

- Minimum confidence.
- Sentiment threshold.
- Maximum failed attempts.
- VIP rules.
- Urgent keyword rules.

- Restricted topics.
- Human availability.
- Fallback destination.
- After-hours behaviour.

Voice

Include:

- Voice selection.
- Speaking speed.
- Warmth.
- Expressiveness.
- Interruption sensitivity.
- Silence handling.
- Background-noise handling.
- Test voice.

Test playground

Create an excellent testing experience where the user can:

- Simulate a phone call.
- Send a WhatsApp message.
- Send an email.
- Start a web chat.
- Inspect the AI response.
- See retrieved knowledge.
- See tool calls.
- See confidence.
- See escalation decisions.
- Mark the response correct or incorrect.

Include draft and published versions of agent settings.

17. KNOWLEDGE BASE

Create a knowledge-management interface.

Supported source cards:

- Website.
- Sitemap.
- Individual URL.
- PDF.
- Document.
- Plain text.
- FAQ.
- Notion.

- Google Drive.
- Manual answer.
- Previous approved conversations.

Each source should show:

- Name.
- Type.
- Status.
- Last synchronized.
- Pages or records indexed.
- Errors.
- Owner.
- Included channels.
- Edit action.
- Resync action.
- Remove action.

Create:

- Knowledge coverage report.
- Conflicting-answer detection.
- Missing-information suggestions.
- Frequently unanswered questions.
- Source preview.
- Search.
- Test question.
- Retrieved citation preview.

Do not imply that uploading a document guarantees accuracy.

Show clear sync and quality states.

18. AUTOMATIONS

Create a polished automation builder.

Users should be able to create rules such as:

- When a new WhatsApp lead arrives, qualify it and notify sales.
- When a call is missed, send a WhatsApp follow-up.
- When an appointment is booked, send email and WhatsApp confirmation.
- When customer sentiment becomes negative, assign a manager.
- When AI confidence is low, require human approval.
- When no response is received for two days, create a follow-up.
- When a call ends, summarize it and update the contact.
- When a new high-value lead appears, send an internal alert.

The automation interface should support:

- Trigger.
- Conditions.
- AI step.
- Action.
- Delay.
- Branch.
- Human approval.
- Test.
- Activity history.
- Enabled or disabled status.

Use a visual flow only if it remains understandable.

Do not build a decorative node canvas that cannot be edited.

19. ANALYTICS

Create analytics for:

Communication

- Conversation volume.
- Channel mix.
- First-response time.
- Resolution time.
- Waiting time.
- Peak hours.
- Missed calls.
- Reopened conversations.

AI

- AI-handled rate.
- Autonomous resolution rate.
- Human takeover rate.
- Average confidence.
- Incorrect-answer reports.
- Tool-action success.
- Knowledge gaps.
- Escalation reasons.

Revenue and leads

- New leads.
- Qualified leads.
- Appointments booked.

- Appointment conversion.
- Lead source.
- Revenue influenced.
- Missed-call recovery.

Team

- Workload.
- Response time.
- Resolution time.
- Availability.
- Conversation ownership.
- Quality review.

Cost

- Phone usage.
- AI usage.
- WhatsApp usage.
- Storage.
- Estimated cost per conversation.
- Estimated cost per appointment.

Create date-range filters and comparison periods.

Do not use misleading chart scales.

20. INTEGRATIONS

Create integration cards for:

Communication

- Gmail.
- Microsoft Outlook.
- WhatsApp Business.
- Website chat.
- Twilio or compatible telephony.
- Existing business phone number.

Calendar and meetings

- Google Calendar.
- Microsoft Calendar.
- Notion.
- Zoom.
- Google Meet.
- Microsoft Teams.

CRM and business

- HubSpot.
- Salesforce.
- Pipedrive.
- Notion databases.
- Webhooks.
- Zapier or compatible automation.
- Generic REST API.

Each card must support realistic states:

- Not connected.
- Connecting.
- Connected.
- Reauthorization required.
- Permission missing.
- Sync delayed.
- Error.
- Disabled.

Provide configuration drawers with fields appropriate to each integration.

Never expose secret tokens after they have been saved.

21. TEAM AND PERMISSIONS

Create team management for:

- Owner.
- Administrator.
- Manager.
- Human agent.
- Analyst.
- Read-only user.

Include:

- Invitations.
- Team availability.
- Skills.
- Departments.
- Languages.
- Working hours.
- Conversation capacity.
- Call availability.
- Assignment rules.

Add role-based access controls for:

- Conversations.
 - Call recordings.
 - Contact exports.
 - AI settings.
 - Knowledge.
 - Integrations.
 - Billing.
 - Audit logs.
-

22. SECURITY AND AUDIT EXPERIENCE

Create visible product interfaces for:

- Two-factor authentication.
- Active sessions.
- Single sign-on placeholder.
- Role-based permissions.
- Data retention.
- Call-recording retention.
- Export requests.
- Deletion requests.
- PII masking.
- IP allowlist placeholder.
- Webhook signing.
- Encryption status.
- Security events.
- Audit log.

The audit log should include:

- Actor.
- Action.
- Resource.
- Timestamp.
- IP or source.
- Result.
- Additional details.

Include audit entries for both human users and AI actions.

23. ONBOARDING

Create a guided onboarding process.

Step 1: Business

Collect:

- Business name.
- Industry.
- Website.
- Time zone.
- Business location.
- Primary use case.

Step 2: Channels

Allow connection of:

- Phone.
- Gmail.
- Outlook.
- WhatsApp.
- Website chat.

Allow skipping while clearly showing what remains incomplete.

Step 3: Knowledge

Allow:

- Website import.
- File upload.
- Notion connection.
- Manual FAQs.
- Business-policy entry.

Step 4: AI identity

Configure:

- Name.
- Voice.
- Tone.
- Greeting.
- Languages.
- Disclosure.
- Handoff.

Step 5: Availability

Configure:

- Working hours.

- Team availability.
- Calendar.
- Appointment types.

Step 6: Test

Run a simulated conversation.

Show:

- AI answer.
- Knowledge used.
- Action taken.
- Confidence.
- Handoff decision.

Step 7: Completion

Display a clear readiness checklist.

Do not claim that the agent is live if required integrations are missing.

24. WEBSITE CHAT WIDGET

Create a separate embeddable website-chat demonstration.

Include:

- Launcher.
- Welcome state.
- AI disclosure.
- Conversation.
- Suggested questions.
- Attachment.
- Appointment booking.
- Human handoff.
- Offline state.
- Consent and privacy link.
- Minimize.
- Close.
- Restart conversation.

The widget should inherit configurable business branding rather than FrontDesk OS branding.

Create a code-snippet installation view, but do not imply the snippet is operational unless the embed route actually works.

25. DEMONSTRATION DATA

Create realistic seeded demonstration data for a professional service business.

Include:

- At least 20 contacts.
- At least 35 conversations.
- At least 8 calls.
- Email conversations.
- WhatsApp conversations.
- Website chats.
- Appointments.
- AI actions.
- Human takeovers.
- Knowledge sources.
- Automation runs.
- Channel errors.
- Audit events.

Avoid lorem ipsum.

Use natural, believable client language.

Include a mixture of:

- New leads.
- Existing clients.
- Urgent issues.
- Routine enquiries.
- Booking requests.
- Dissatisfied customers.
- Spam.
- AI uncertainty.
- Successful AI resolution.

Do not fabricate testimonials or business success claims.

26. DATA TYPES

Create strict TypeScript models for at least:

- Workspace.
- User.
- Team.
- TeamMember.
- Role.
- Integration.

- Channel.
- Contact.
- ContactIdentity.
- Company.
- Conversation.
- Message.
- Attachment.
- Call.
- CallTranscriptSegment.
- Appointment.
- AvailabilityRule.
- KnowledgeSource.
- KnowledgeDocument.
- Automation.
- AutomationRun.
- AIConfiguration.
- AIResponse.
- AIAction.
- Handoff.
- UsageRecord.
- Notification.
- AuditLog.

Create adapter interfaces for future production integrations.

Examples:

- Email adapter.
- WhatsApp adapter.
- Voice adapter.
- Calendar adapter.
- Notion adapter.
- AI provider adapter.
- Storage adapter.

The UI should consume these interfaces rather than depending directly on one vendor.

27. ENVIRONMENT AND INTEGRATION STUBS

Create a complete `.env.example` using placeholder values only.

Possible categories:

```
APP_URL=
DATABASE_URL=
AUTH_SECRET=

GOOGLE_CLIENT_ID=
```

```
GOOGLE_CLIENT_SECRET=

MICROSOFT_CLIENT_ID=
MICROSOFT_CLIENT_SECRET=

WHATSAPP_ACCESS_TOKEN=
WHATSAPP_PHONE_NUMBER_ID=
WHATSAPP_VERIFY_TOKEN=

TWILIO_ACCOUNT_SID=
TWILIO_AUTH_TOKEN=
TWILIO_PHONE_NUMBER=

NOTION_CLIENT_ID=
NOTION_CLIENT_SECRET=

AI_PROVIDER_API_KEY=

STORAGE_ACCESS_KEY=
STORAGE_SECRET_KEY=

SENTRY_DSN=
```

Never commit real secrets.

Never expose server secrets through client environment variables.

Mock adapters must continue to make the interface usable without credentials.

28. UX STATES

Every major module must include:

- Loading state.
- Empty state.
- Error state.
- Disconnected state.
- Permission-denied state.
- No-results state.
- Offline state.
- Expired-authentication state.
- Partial-sync state.
- Rate-limit state.
- Successful action state.

Include realistic failures such as:

- WhatsApp template unavailable.
- Gmail authorization expired.

- Calendar conflict.
- Call transfer failed.
- AI confidence too low.
- Knowledge source failed to sync.
- Attachment too large.
- Human agents unavailable.

A professional product is defined partly by how it handles failure.

29. MOTION AND INTERACTION

Translate the reference website's motion language into a SaaS interface.

Use motion for:

- Page entrance.
- Sidebar collapse.
- Conversation selection.
- New message arrival.
- AI listening.
- AI speaking.
- Call transfer.
- Appointment creation.
- Drawer transitions.
- Modal transitions.
- Filter changes.
- Chart updates.
- Toast notifications.

Do not animate every element.

Do not delay core interactions for decorative effects.

Respect reduced-motion preferences.

Use transforms and opacity wherever possible.

Avoid scroll-jacking.

30. RESPONSIVE REQUIREMENTS

Test at:

- 1440px.
- 1280px.
- 1024px.
- 768px.

- 430px.
- 390px.
- 360px.

Requirements:

- No horizontal page overflow.
- No hidden primary actions.
- No unreadable tables.
- No overlapping drawers.
- No clipped dropdowns.
- No off-screen modals.
- No tiny touch targets.
- No navigation covering content.
- No fixed elements hiding the composer.
- No broken calendar layout.
- No dashboard cards with overflowing text.

Convert complex tables into mobile cards where appropriate.

31. ACCESSIBILITY

Meet strong WCAG practices.

Include:

- Semantic HTML.
- Keyboard navigation.
- Visible focus states.
- Accessible labels.
- Correct dialog focus management.
- Proper table markup.
- Meaningful alt text.
- Sufficient colour contrast.
- Status information that is not colour-only.
- Reduced-motion support.
- Screen-reader announcements for important live events.

The inbox and call controls must be usable without a mouse.

32. PERFORMANCE

Requirements:

- Avoid unnecessary client components.
- Lazy-load heavy charts and secondary modules.
- Avoid re-rendering the complete inbox for one message.

- Use optimized fonts.
 - Use optimized images.
 - Avoid large decorative video backgrounds.
 - Keep initial marketing-page JavaScript controlled.
 - Do not load every application module on the homepage.
 - Prevent layout shift.
 - Provide skeleton states.
 - Keep animation smooth on ordinary laptops and phones.
-

33. README

Create a professional `README.md`.

It must include:

1. Product overview.
2. Product positioning.
3. Screenshots.
4. Major features.
5. Route map.
6. Technology stack.
7. Architecture overview.
8. Data model overview.
9. Local installation.
10. Windows PowerShell 5.1 setup commands.
11. Environment variables.
12. Demo mode explanation.
13. Integration-adapter explanation.
14. Available scripts.
15. Testing.
16. Deployment.
17. Security considerations.
18. Current limitations.
19. Production roadmap.
20. Contribution and ownership information.

Use PowerShell 5.1-compatible commands.

Do not use `&&` in PowerShell instructions.

Do not include fake deployment information.

Do not claim real provider integrations unless they were actually implemented and tested.

34. GITHUB REPOSITORY

Create a new GitHub repository under:

witejackel-eng

Repository name:

ai-frontdesk-inbox

Description:

Omnichannel AI front desk and shared client inbox for phone calls, email, WhatsApp, website chat and appointment scheduling.

Requirements:

- Initialize Git.
 - Add a correct .gitignore.
 - Add .env.example.
 - Do not commit secrets.
 - Do not add a permissive open-source license unless explicitly required.
 - Create meaningful commits.
 - Keep commit authorship under witejackel-eng.
 - Do not add AI co-authors.
 - Do not add bot attribution.
 - Do not commit screenshots containing personal credentials.
 - Push the completed code.
 - Verify the default branch.
-

35. TESTING

Run:

- Dependency installation.
- Lint.
- Type checking.
- Production build.
- Unit tests.
- Critical component tests.
- Responsive browser tests.
- Basic accessibility checks.
- Navigation tests.
- Form validation tests.
- Inbox interaction tests.
- Modal and drawer tests.
- Mobile tests.

Verify:

- Every route renders.
- Every navigation item works.

- No missing imports.
- No console errors.
- No hydration warnings.
- No broken images.
- No dead buttons.
- No TypeScript errors.
- No obvious accessibility violations.
- No exposed environment secrets.

Fix failures before finishing.

Do not merely report failures that you introduced.

36. CONTENT RULES

Do not use:

- "Revolutionize."
- "Seamlessly."
- "Cutting-edge."
- "Game-changing."
- "Next-generation."
- "Transform your business."
- Empty AI buzzwords.
- Unsupported performance claims.
- Fake customer counts.
- Fake testimonials.
- Fake partner logos.
- Fake awards.

Write clear product copy that explains:

- What happens.
 - Who benefits.
 - What remains under human control.
 - What requires an integration.
 - What is currently demonstration data.
-

37. COMPLETION STANDARD

The task is complete only when:

- The reference visual language has been extracted and applied.
- The marketing site is complete.
- Authentication screens exist.
- Onboarding is complete.
- The application shell is complete.

- Unified inbox is complete.
- Phone interface is complete.
- Contacts are complete.
- Calendar is complete.
- AI Front Desk builder is complete.
- Knowledge base is complete.
- Automations are complete.
- Analytics are complete.
- Team management is complete.
- Integrations are complete.
- Settings are complete.
- Billing interface is complete.
- Security interface is complete.
- Audit log is complete.
- Mobile layouts work.
- Demo data is realistic.
- Empty and error states exist.
- README is complete.
- Repository is created and pushed.
- Lint passes.
- Type checking passes.
- Production build passes.
- Critical interactions have been tested.

Do not stop after producing several attractive screens.

Do not return a plan as the final result.

Implement the product.

38. FINAL RESPONSE

At completion, provide:

1. Concise implementation summary.
2. GitHub repository location.
3. Deployment location if deployed.
4. Routes implemented.
5. Tests executed.
6. Test results.
7. Integrations operating in demo mode.
8. Environment credentials still required.
9. Any genuine limitations.
10. Recommended next production-development step.

Do not exaggerate what was completed.